

Kalp Shah

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EXPERIENCE

Energy Toolbase

Software Engineer

Calgary, AB

Jan. 2021 – Present

- Collaborated in an 8–10 engineer core team building the flagship edge Energy Management System (EMS) deployed across **500+ commercial sites (100+ MWh capacity)**, writing high-frequency networking protocols in **Go** and core dispatch algorithms in **Python**.
- Led a team of 3 engineers building an edge-to-cloud telemetry ingestion pipeline on **AWS** (IoT Core, Kinesis, Lambda) and **PostgreSQL**, streaming **10M+ daily records** across 500+ commercial sites with zero downtime.
- Architected an enterprise cloud Weather API proxy service in **FastAPI** and **PostgreSQL** on **Kubernetes**, serving **5M+ daily requests** with 99.99% uptime and sub-50ms latency via intelligent upstream caching.
- Orchestrated multi-service edge runtimes with **Docker Compose** over a local **MQTT bus**; downsampled sensor telemetry to 10 seconds in **InfluxDB**, cached live state in **Redis**, and buffered streams to AWS via **IoT Greengrass**.
- Architected secure remote device access and bidirectional control channels across 500+ field gateways using **WireGuard VPN** and **REST APIs** for remote setpoint adjustments and diagnostics.
- Overhauled battery dispatch algorithms using **Linear Programming in Python (NumPy, Pandas)**, replacing static heuristics with dynamic mathematical optimization to maximize customer energy savings.
- Conducted technical design reviews, mentored 4 junior engineers on systems programming, and expanded automated test coverage (**TDD**) to over **85%**.

Celestica

Test & Automation Developer

Newmarket, ON

May 2018 – Sept. 2019

- Collaborated in a 12-engineer team to build automated test benches and Python verification suites for enterprise network routers for a tier-1 US telecommunications client in a high-volume manufacturing facility.
- Engineered automated Python regression test suites and fault-injection scripts with a Design-for-Test (**DFT**) methodology to identify hardware and firmware flaws early in the production lifecycle.
- Streamlined manufacturing verification cycles, reducing manual inspection overhead and accelerating test throughput on factory lines.

PROJECTS

Second Brain: AI Knowledge & Retrieval Platform | *Python, FastAPI, Qdrant, Ollama, Docker*

- Built a private, offline AI knowledge base (RAG) indexing technical documents and code, pairing semantic vector search (**Qdrant**) with exact keyword matching (**BM25**).
- Implemented a two-stage retrieval pipeline with cross-encoder reranking, increasing retrieval accuracy by **35%** while streaming local model responses (**Ollama**) in real time via **FastAPI** and **SSE**.

Personal Cloud Platform & Infrastructure | *Docker Compose, Cloudflare Tunnels, systemd, Bash, Python*

- Architected and deployed self-hosted multi-service production infrastructure (kalp.dev) featuring zero-trust ingress via **Cloudflare Tunnels** with no open inbound ports.
- Developed automated disaster recovery pipelines with **systemd** timers, **Bash**, and offsite encrypted storage, validated by Python integration test suites.

EDUCATION

Ryerson University

Bachelor of Engineering (B.Eng.) in Computer Software Engineering

Toronto, ON

TECHNICAL SKILLS

Languages: Python, Go (Golang), SQL (PostgreSQL), Bash / Shell Scripting

IoT & Hardware Protocols: Modbus (RTU/TCP), CANbus (SocketCAN), MQTT, Industrial Edge Controllers, Embedded Linux, systemd

Cloud & DevOps: AWS (IoT Greengrass, IoT Core, Kinesis, Lambda, EC2, S3), Docker Compose, Kubernetes, WireGuard, Datadog, SaltStack

Databases & Frameworks: PostgreSQL, Redis, InfluxDB, Qdrant (Vector DB), FastAPI, REST APIs, Microservices, TDD/BDD